Normal Distributions and the Empirical Rule

Name _____

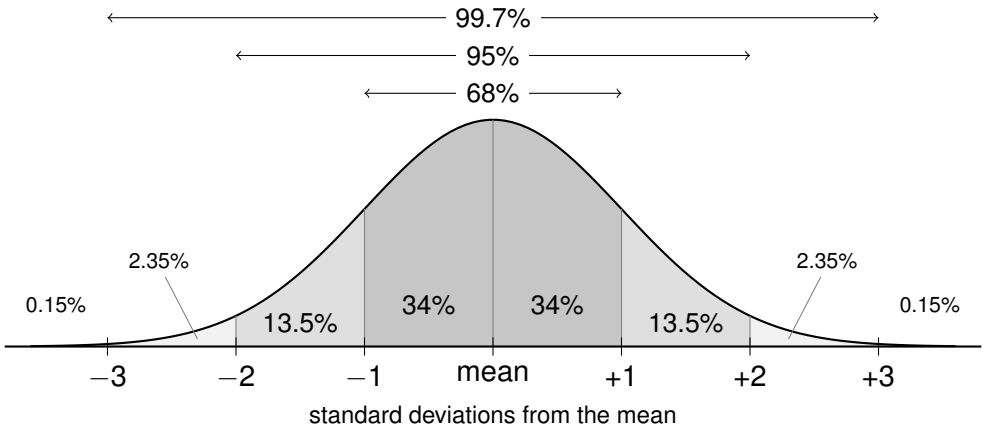
Measure the heights of thousands of adult women. The histogram makes a smooth **bell shape**. Data like this has a **normal distribution**.

A normal distribution:

- is bell-shaped and symmetric
- has the mean and the median in the center (they are equal)
- is described by two numbers: the **mean** μ (center) and the **SD** σ (spread)

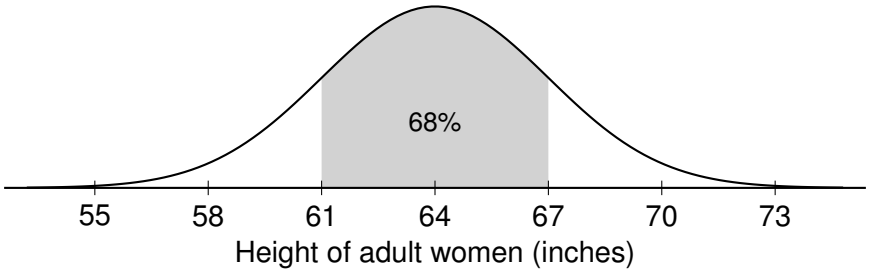
A larger SD makes the curve wider and flatter. A different mean moves the curve left or right.

The empirical rule (68-95-99.7 rule)



About **68%** of the data is within **1 SD** of the mean.
About **95%** of the data is within **2 SDs** of the mean.
About **99.7%** of the data is within **3 SDs** of the mean.

Example: Heights of adult women have a mean of about 64 inches and an SD of about 3 inches.



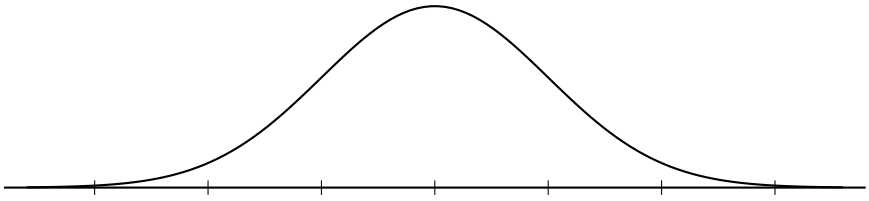
Percent from 61 to 67 inches	68% (within 1 SD)
Percent from 58 to 70 inches	95% (within 2 SDs)
Percent taller than 67 inches	13.5% + 2.35% + 0.15% = 16%
Percentile of 67 inches	100% − 16% = 84%, the 84th percentile

Tip: Draw the curve and label it every time. Then add up the parts you need.

Work with a partner. Compare answers when you finish each problem.

1. The weights of apples from one orchard have a normal distribution. The mean is 150 grams and the SD is 10 grams.

   Label the number line with the weights.



Weight (grams)

   About what percent of the apples weigh:

   (a) from 140 to 160 grams?

   (b) from 130 to 170 grams?

   (c) more than 160 grams?

   (d) less than 130 grams?

   (e) from 150 to 170 grams?

   (f) from 130 to 160 grams?

2. A phone battery lasts a mean of 20 hours with an SD of 2 hours. The times have a normal distribution.

   (a) Between what two times do the middle 95% of batteries last?

   (b) What percentile is 22 hours?

   (c) About what percent of batteries last less than 16 hours?

   (d) A battery lasts 27 hours. Is that unusual? Explain.

3. In a normal distribution, what percentile is a value with each z-score?

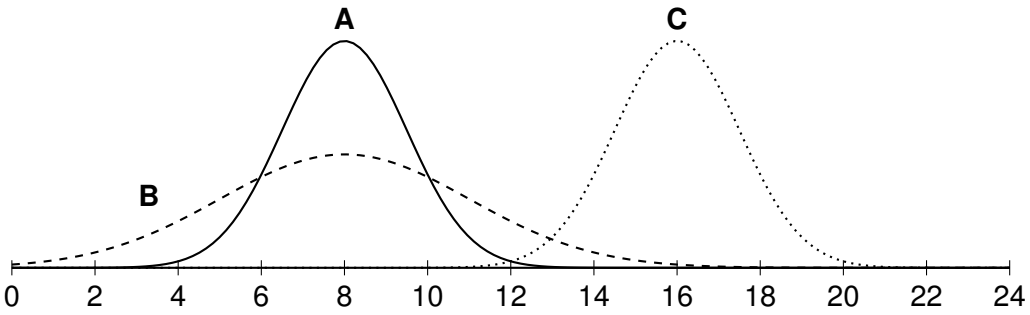
   (a) $z = 0$

   (b) $z = 1$

   (c) $z = -2$

   (d) $z = 2$

4. The three curves are normal distributions.



   (a) Which two curves have the same mean?

   (b) Which curve has the largest SD? Explain.

   (c) How are A and C the same? How are they different?

5. Would each data set have a normal distribution? Explain.

   (a) The heights of all adult men in Missouri.

   (b) The yearly income of every household in the US.

   (c) Scores on a very easy test.